

PRODUCT, MOBILE AND DELIVERY BLUEPRINT

XELOR Unified Factory and Supply Network

A combined solution inspired by IndiaMART's supplier reach and Vyapar's everyday simplicity - extended with XELOR's factory-grade planning, production, quality, maintenance, finance and governance.

SOURCE	RUN	SEE	COLLECT	PROVE
Find supply	Run the factory	See it by phone	Bill and collect	Keep evidence

DECISION DOCUMENT | VERSION 1.0 | 30 AUGUST 2026
Prepared for AIKYANTRA - XELOR

1. Executive decision

The recommended upgrade is one connected industrial operating system with five focused surfaces. It should not become a clone of IndiaMART or a generic billing application.

Recommended product statement

XELOR connects supplier discovery, structured sourcing, factory execution, commercial documents, cash visibility and role-safe mobile access in one governed loop.

IndiaMART lesson	Vyapar lesson	XELOR advantage
Broad supplier discovery, location-aware search, RFQs, quick contact and marketplace liquidity.	Fast billing, simple stock/accounting, mobile access, offline resilience, WhatsApp/PDF sharing and owner-friendly language.	MRP (what material is needed and when), governed purchasing, inventory integrity, production, quality, maintenance, finance and evidence-backed decisions.

The combined promise



Defensible difference

IndiaMART answers who claims they can supply. Vyapar answers what the business billed, bought and owes. XELOR should answer which qualified supplier can meet the exact requirement, how it affects the factory plan, and what operational, quality and financial evidence proves the outcome.

What this blueprint decides

- Build **XELOR Source** as a curated RFQ-led industrial supplier network, starting invite-only.
- Build **XELOR Flow** for quotations, branded documents, receivables, supplier invoices and matching.
- Build **XELOR Pocket** as an installable, task-first mobile PWA rather than shrinking the whole desktop ERP.
- Keep XELOR Core authoritative for planning, stock, quality, approvals and accounting.
- Start with field validation in one industrial cluster and two or three repeat-demand categories.

Evidence: [IM-1], [IM-2], [VY-1], [VY-2], [XE-1], [XE-2]. Full references appear in Appendix B.

2. Contents and decision map

This document moves from product strategy to workflows, mobile design, architecture, phased delivery, operating metrics and risk controls.

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How to use this document

Sections 3-9 define the product and operating flows. Sections 10-14 define the mobile solution. Sections 15-17 define architecture, data and repository changes. Sections 18-25 define groundwork, delivery, metrics and controls. Section 26 gives the recommendation.

Decision status

Decision	Recommendation	Status
Marketplace model	Curated supplier network plus structured RFQ	Proceed to discovery
Mobile channel	Installable PWA first; native only after evidence	Recommended
Data model	Private tenant, published network and participant transaction zones	Non-negotiable
First geography	One dense industrial cluster	Select in Phase 0
First categories	Two or three repeat-demand industrial categories	Validate in field
Payments/logistics	Partner or defer until transaction evidence exists	Deferred

3. What to take from IndiaMART and Vyapar

The goal is selective inspiration: retain the mechanisms that reduce friction, improve the weak controls, and avoid copying business models that do not fit a factory ERP.

Dimension	IndiaMART	Vyapar	Improved XELOR
Primary job	Find suppliers and create business enquiries	Bill, track stock and manage basic accounts	Source, run, see and control the industrial loop
Buyer experience	Search, filters, supplier profile, contact or RFQ	Add party/items, create document, share and collect	MRP need or enquiry becomes a governed transaction
Trust	GST, TrustSEAL, ratings and limited payment protection	User-entered records, backups and audit/accounting reports	Evidence-specific verification plus transaction-derived supplier performance
Mobile	Buyer/seller marketplace app	Core business activity on phone and desktop	Role-specific factory pulse, tasks, capture and approvals
Manufacturing	Marketplace listings and job work	Basic BOM recipe and consumption	Multi-level planning, production, quality, maintenance and traceability
Main weakness	Noisy listings, inconsistent data and off-platform fulfilment	Crowded forms, hidden settings and light factory controls	Must simplify without weakening governance

Evidence that matters

IndiaMART scale	Vyapar adoption	Strategic relationship
At 30 June 2026 IndiaMART reported 234M registered buyers, 8.8M supplier storefronts, 218K paying suppliers and 132M listings. Scale creates discovery, but not guaranteed fulfilment.	Google Play reports 10M+ downloads and a 4.8 rating. Its value is low-friction daily administration for small businesses, not factory-grade control.	IndiaMART reported a 28.59% fully diluted stake in Simply Vyapar Apps at 31 March 2026. They remain separate products with complementary roles.

Copy, improve, avoid

Copy	Improve	Avoid
Search, RFQ and supplier reach	Use category-specific technical schemas and exact revisions	A generic catalogue claiming every item fits
Fast invoice/quote creation	Make important factory documents quick without losing approval	One enormous form with every possible option
WhatsApp/PDF convenience	Share authenticated links and audited documents	Sending confidential drawings through open messages
Phone visibility	Task-first views, evidence and freshness timestamps	Putting 24 desktop modules in a small menu
Freemium-style onboarding lessons	Guided setup and Excel import	Entering a mass-market price war with retail billing tools

Evidence: [IM-1], [IM-2], [IM-3], [VY-1], [VY-2], [VY-3], [VY-4].

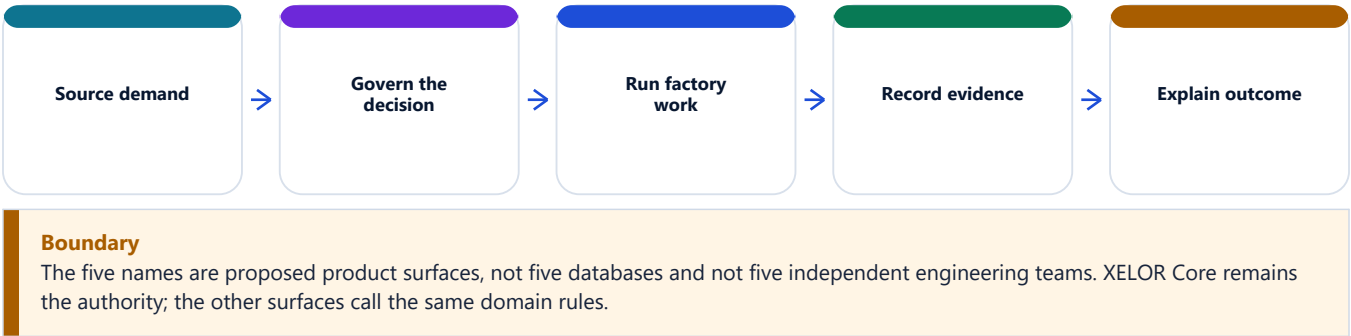
The scale and adoption figures above are sourced facts. The weakness rows are product-analysis hypotheses and must be validated with buyers and suppliers during Phase 0.

4. The proposed product system

Use five connected surfaces. They share identity, permissions, audit, workflow, data and events, but each presents only the work its user needs.

XELOR Core The existing factory system of record: engineering, MRP, purchase, inventory, production, quality, maintenance, sales and accounts.	XELOR Source Supplier discovery, verification, structured RFQs, quote revisions, comparison, award and supplier performance.	Supplier Connect A lightweight portal for suppliers who do not use XELOR ERP: profile, catalogue, RFQ, quote, PO acknowledgement and delivery promise.
XELOR Flow Quotations, commercial documents, GST-ready PDFs, receivables, supplier invoices, matching and reminders.	XELOR Pocket Curated mobile visibility, alerts, approvals, scan/capture and assigned work for factory and commercial roles.	

One truth, several experiences



Positioning

XELOR Unified: the industrial operating system that connects customer demand, qualified supply, factory execution, commercial documents and management visibility from desktop and phone.

A simple external promise can be: **Source. Run. See. Collect. Prove.** The deeper proof is the audit trail from requirement through material, work, quality and money.

Commercial boundaries

Surface	Initial commercial model	Integrity rule
Core + Flow + Pocket	Factory subscription with site and role tiers	No loss of audit, approval or ledger controls at a lower tier
Source - buyer	Subscription or usage tier validated in pilot; avoid pay-per-lead incentives	Ranking and award remain evidence-led
Supplier Connect	Free invited RFQ response and basic profile; optional paid services later	Payment never changes qualification, score or award recommendation

5. People and jobs to be done

The combined product must serve different roles without giving every person access to every record.

Persona	Question they need answered	Primary surface	Phone role
Owner / Managing Director	Are promises, production and cash under control?	Pocket + Flow	See exceptions, approve, escalate
Plant manager	What is late, stopped, blocked or rejected today?	Pocket + Core	Shift pulse and interventions
Planner	What must be made or bought, and by when?	Core + Source	Shortage and arrival risk
Buyer	Who can meet the exact need at acceptable total risk?	Source	RFQs, clarification, shortlist
Purchase manager	Is the award commercially and operationally defensible?	Source + Pocket	Evidence-led approval
Production supervisor	What should run next and what is blocking it?	Core + Pocket	Work queue and counts
Operator	What is my next job and what must I record?	Pocket	Scan, start, quantity, stop reason
Stores user	What arrived, where does it go, and is it valid?	Core + Pocket	Scan and prepare GRN
Quality inspector	What must be tested and what evidence determines disposition?	Core + Pocket	Readings, photos, result
Maintenance technician	What asset needs work and what restores it safely?	Core + Pocket	QR work order and checklist
Sales / accounts	What should be quoted, billed, collected or paid?	Flow	Share and follow up
Supplier representative	What is required and how do I respond?	Supplier Connect	Quote, acknowledge, update promise

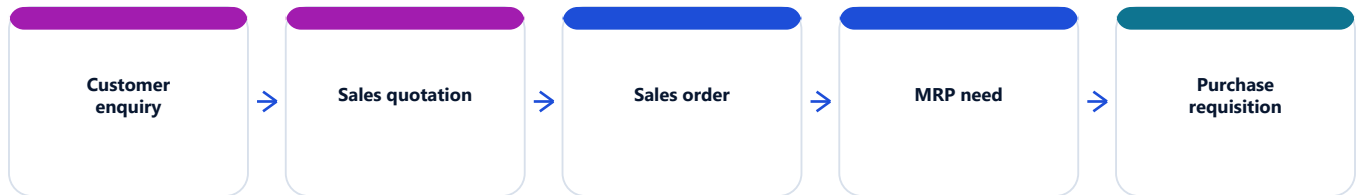
Permission principle

The owner may want broad visibility, but mobile still respects tenant, plant, role and field rules. Operators do not see margins; suppliers never see the buyer's internal ERP; sensitive values stay behind explicit permission.

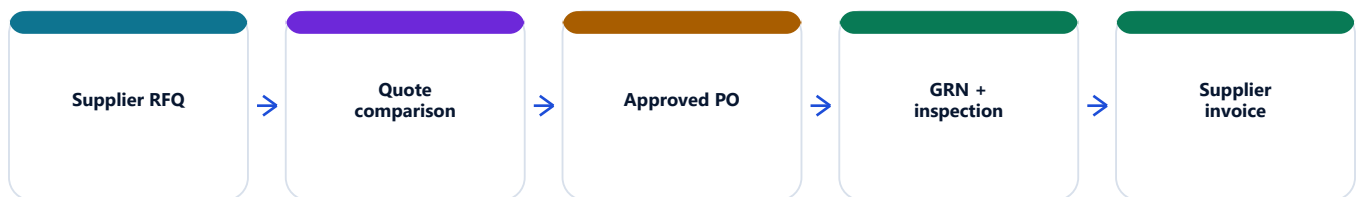
6. North-star closed loop

This is the single story the combined solution must complete without spreadsheets or retyping.

Demand and planning



Source to receive



Make to cash and evidence



Why this is stronger than a marketplace alone

- The requirement comes from a real customer promise or material plan, not an unqualified enquiry.
- The RFQ preserves item, quantity, need date, drawing revision, plant and planning evidence.
- The award remains human-approved and creates the existing draft PO exactly once.
- The receipt and inspection prove whether the supplier actually delivered acceptable material.
- The supplier score comes from immutable transaction outcomes rather than anonymous stars.
- The commercial outcome connects to invoice, receivable, payable and cash evidence.

Golden rule

MRP may recommend a purchase need. It must never publish confidential demand, select a supplier or commit money without the governed human boundary.

7. XELOR Source: buyer and supplier journeys

Start with invited suppliers and known industrial categories. Public discovery can expand only after catalogue quality, access control and transaction evidence are proven.

Buyer journey

- 1 Open an MRP shortage or create a manual sourcing requirement.
- 2 Confirm item, quantity, UOM, delivery plant, need date and confidentiality.
- 3 Attach a controlled drawing revision, tolerances, tests and certification needs.
- 4 Search approved vendors first, then published network suppliers by capability and location.
- 5 Invite a shortlist; contacts stay masked until the configured consent point.
- 6 Receive immutable quote revisions containing price, tooling, tax, freight, MOQ, capacity, validity, lead time and deviations.
- 7 Compare raw facts and an explainable policy score; record the recommendation reason.
- 8 Approve the award under separation-of-duties policy and create the existing draft PO through an idempotent writer plus a database-unique award-to-PO key.

Supplier journey

- 1 Create a lightweight organisation and verify identity, GST and selected evidence.
- 2 Publish only chosen capabilities, locations, certifications and catalogue records.
- 3 Accept or decline an RFQ invitation; view only the shared package.
- 4 Submit or revise a quotation without overwriting earlier revisions.
- 5 Respond to structured clarifications inside the RFQ record.
- 6 Acknowledge an awarded PO or propose a revised promise for buyer review.
- 7 Submit dispatch details and later see the supplier's own receipt and quality outcomes.

Marketplace state model

Object	Recommended states	Control
RFQ	Draft -> Published -> Evaluation -> Awarded / Closed / Cancelled	Only buyer can change lifecycle
Quote	Draft -> Submitted -> Revised / Withdrawn / Expired	Supplier owns submissions; revisions immutable
Award	Proposed -> Pending approval -> Approved / Rejected -> Converted	Workflow approval plus idempotent PO conversion
Supplier profile	Draft -> Submitted -> Verified -> Published -> Expired / Suspended	Evidence-specific verification and freshness

8. RFQ quality, comparison and supplier trust

A low price is not automatically a good industrial decision. XELOR must make technical deviations, arrival feasibility and quality history visible before any score.

RFQ minimum package

Technical Part code, category attributes, material grade, tolerances, drawing revision, substitutes and test method.	Quantity and time Quantity, UOM, pack multiple, need date, quote deadline, delivery location and schedule split.	Commercial Taxes, freight ownership, tooling/NRE, payment assumptions, warranty and validity.
Quality Inspection plan, certificates, sampling, traceability, rejection policy and corrective action.	Confidentiality Classification, invited participants, NDA state, file expiry, watermark and access audit.	Evidence Originating demand, requisition, plant, planner, approvals and reason for source selection.

Comparison view

Factor	What must be shown	Illustrative weight
Landed cost	Unit price + tooling + freight + non-creditable tax	30%
Delivery feasibility	Confirmed dispatch, arrival date, MOQ and capacity	25%
Quality history	Accepted quantity, rejection, returns and CAPA closure	20%
Delivery history	On-time-in-full and promise adherence	15%
Compliance	Required certificates, capability and documentation	10%

Technical gate before scoring

The buyer first marks each response Pass, Conditional or Fail against the released specification. Failed quotes are excluded. Conditional deviations and equivalent parts require recorded engineering acceptance. Missing history is Unrated, not zero. Start with one-supplier awards; add split awards only after allocation controls are proven. Then show raw facts, explain every weighted score and require a separate authorised approver above policy thresholds; the supplier-profile verifier cannot approve that supplier's award.

Verification must be specific

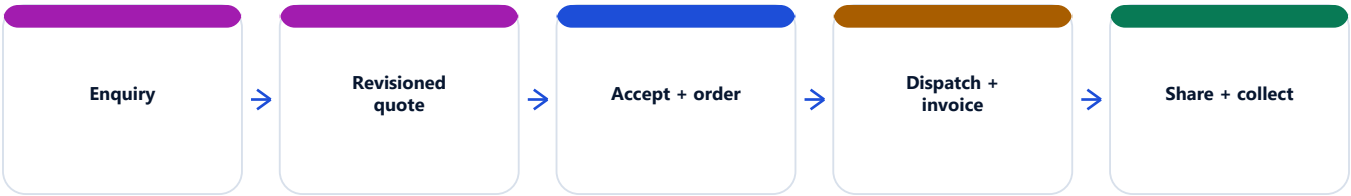
Replace one vague badge with separate statements: legal identity verified; GST verified; bank account verified; factory visited; capability evidence reviewed; certificate valid until date; buyer reference checked; XELOR transaction history available; profile last updated on date.

Capture immutable delivery and inspection events first. Publish a supplier score only after a defined sample threshold; otherwise show **Insufficient evidence**. Version the formula, disclose its window and confidence, and provide a supplier correction/dispute route. Sponsored visibility must never affect conformity, scoring or award recommendation.

9. XELOR Flow: Vyapar-inspired commercial simplicity

XELOR already owns the order, dispatch, customer invoice and receipt spine. The upgrade should close the quotation, document, payables and everyday usability gaps without replacing working controls.

Quote to cash



Requisition to pay



Target capabilities across phases

- Customer enquiry and revisioned quotation with validity, tax, freight, payment terms and delivery promise.
- Branded quote, PO, delivery challan, invoice, credit note, debit note and receipt PDFs.
- Audited email and approved WhatsApp delivery using authenticated links for sensitive content.
- One-click quote-to-order conversion with no item or term retyping.
- Receivables ageing, reminder schedules, collection worklist and party statement.
- Supplier invoice capture with duplicate constraints, configurable match tolerances, tax/ITC treatment, PO-GRN-invoice comparison, AP open items and ageing.
- Encrypted and masked bank masters; independently verified bank changes with cooling period; maker-checker payment approval; signed bank/API handoff; settlement import, allocation, reversal and reconciliation.
- Fast item/customer creation with duplicate checks and controlled Excel onboarding.

Simplicity standard

Vyapar pattern	XELOR implementation
Visible Add Sale / Add Purchase	Five role-aware daily actions, not module hunting
Automatic stock/accounting updates	Existing domain services continue to post atomically
Share via WhatsApp/PDF	Branded documents, audit and secure link policy
Owner-friendly reports	Plain-language cash, order, stock and risk views backed by ledger data
Offline billing	Bounded draft capture; authoritative stock/finance remains server-confirmed

Evidence: [VY-1], [VY-2], [VY-3], [XE-2], [XE-4].

10. XELOR Pocket: the factory in your phone

The phone must answer what is happening now, what is at risk, what work is mine, what decision is waiting and what I can safely record where I am standing.

Product decision

Build a dedicated installable mobile PWA. Do not compress the desktop ERP. Use the same identity and domain services, but a task-first information architecture and low-bandwidth read model.

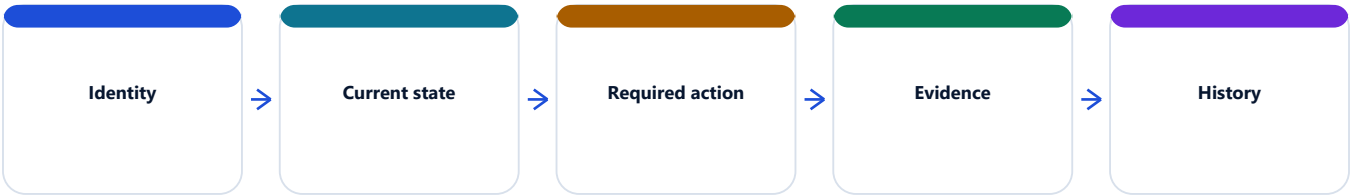
Five destinations

Home Role-specific factory pulse and today's priorities.	My Work Assigned operations, inspections, receipts, maintenance, RFQs and approvals.	Scan Identify machine, item, batch, bin, PO, work order or inspection.
Alerts Critical, urgent and attention items with evidence and ownership.	More Role-permitted secondary areas, settings, help and sign-out.	

Persistent context

Every mobile screen shows company, plant, shift where relevant, online/offline state, last synchronisation time and number of queued drafts. Every metric shows its source and freshness state: Fresh, Stale or Unknown.

Record detail order

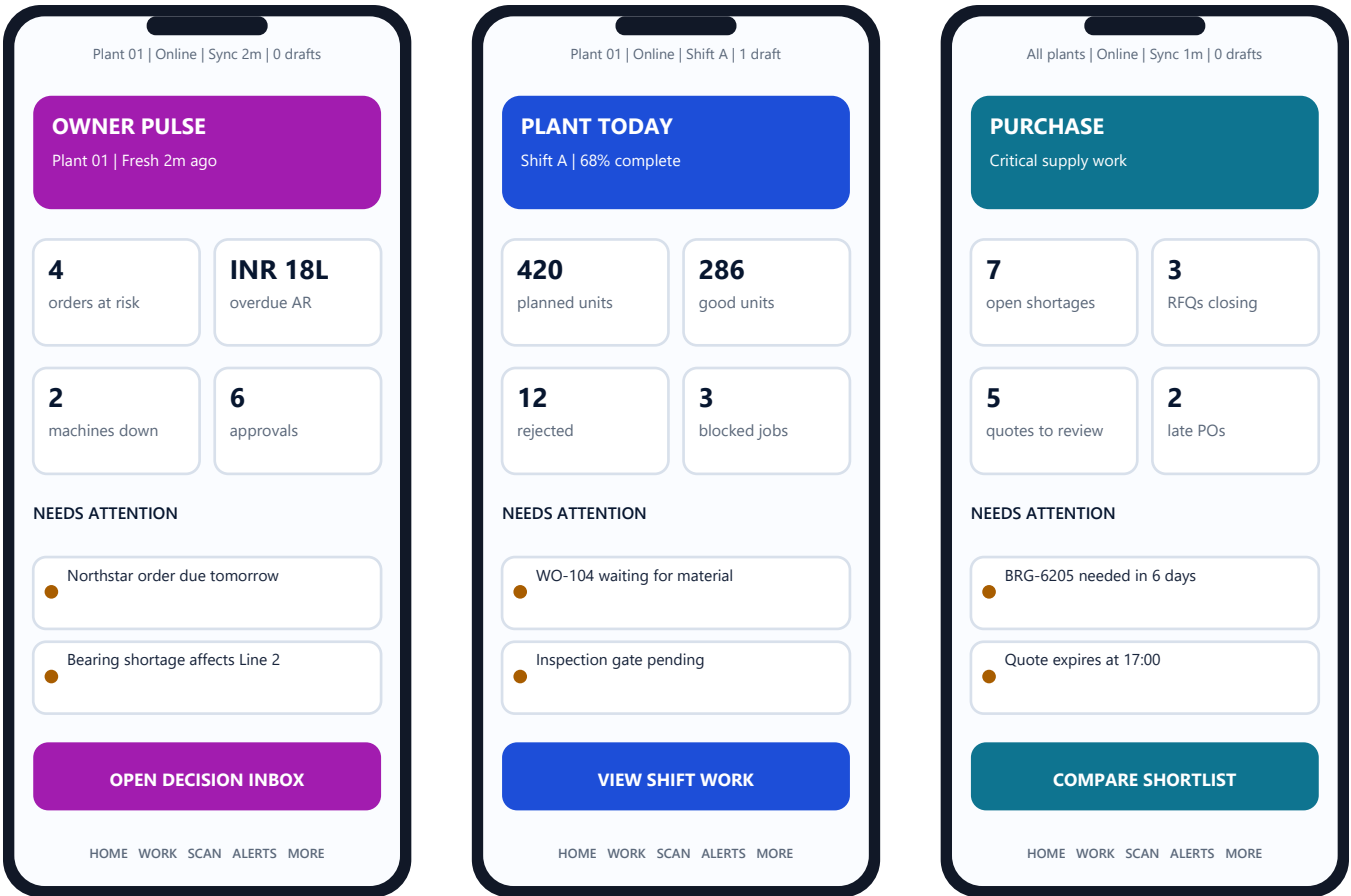


No false live status

A number without a fresh accepted source is Stale or Unknown. The phone must never show a machine, order or material condition as green merely because the last value was green.

11. Mobile role dashboards

The following conceptual screens show how different users can see the same factory truth without receiving the same access.



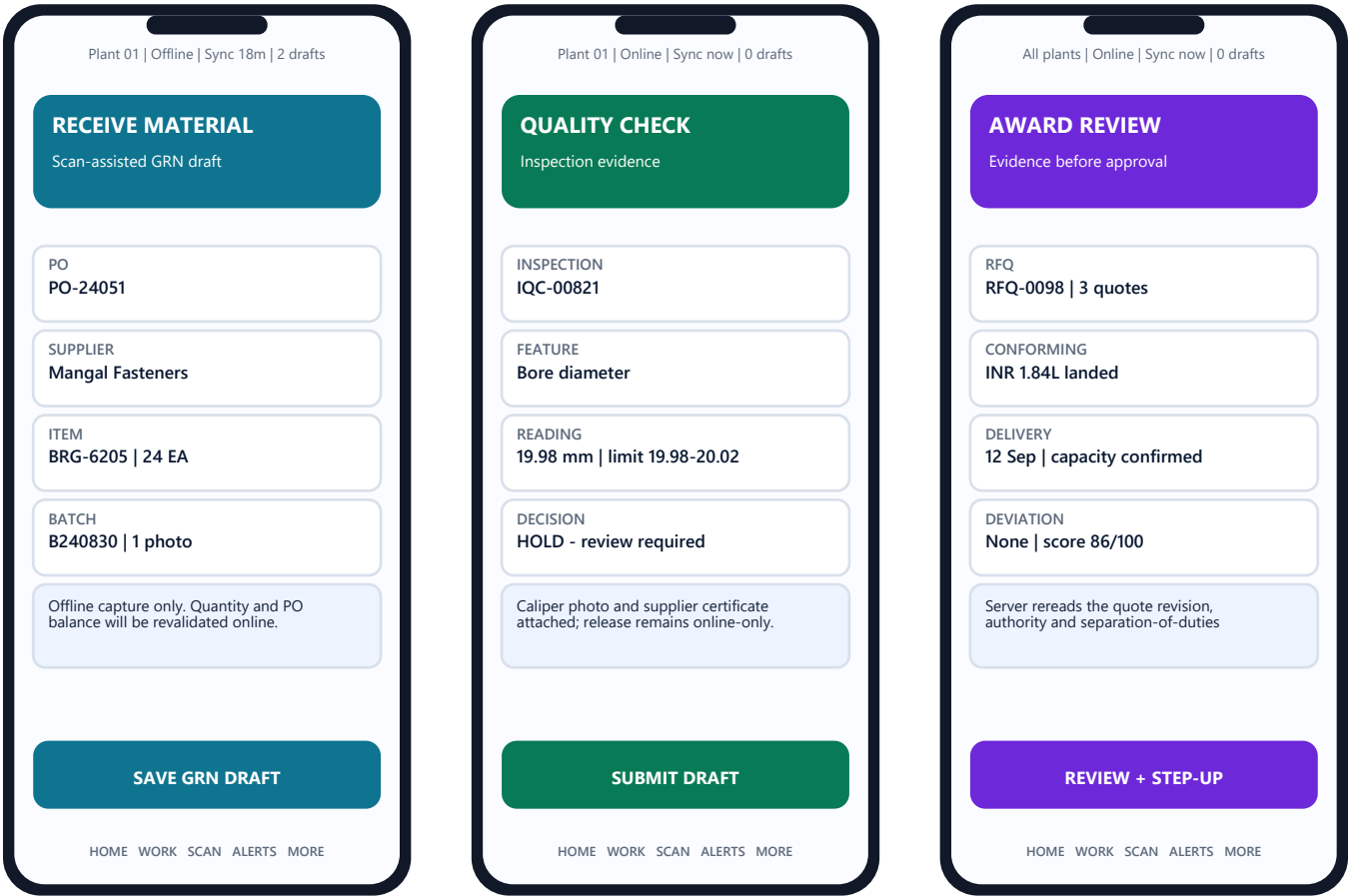
Conceptual mobile screens. Values are illustrative; production screens must read current authorised records and show the source timestamp.

Owner mobile outcome

An owner should understand customer promise risk, production progress, shortages, quality, downtime, dispatch, cash and pending approvals without calling each department. The app should prioritise exceptions and trends, not reproduce every transaction.

Mobile action screens: capture and approval

These concepts demonstrate that Pocket is more than a dashboard. It supports point-of-work evidence and governed decisions while keeping authoritative posting on the server.



Visible state contract

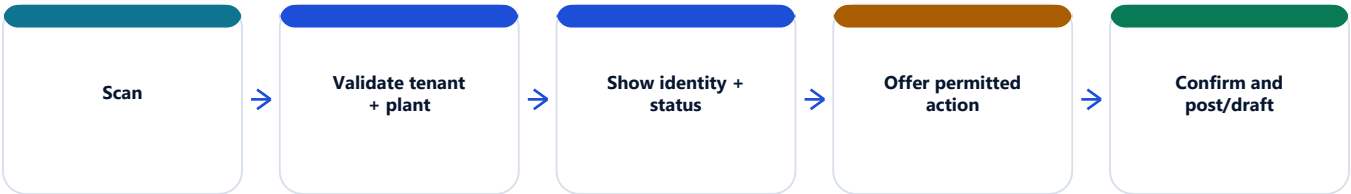
Each screen shows organisation or plant, connectivity, sync age and queued drafts. Bottom navigation remains stable. Offline work is visibly Draft; an approval always reloads authoritative evidence before committing.

12. Mobile work on the factory floor

The phone becomes a controlled point-of-work tool for capture and evidence. It does not become a safety controller or an unrestricted transaction terminal.

Role	See	Safe mobile actions	Online-only commitments
Production supervisor	Queue, material readiness, progress, rejects, downtime	Assign operator; start/complete operation; record counts; hold with reason	Final production posting after validation
Operator	Next job, item, quantity, machine, instruction revision	Scan, start, good/reject quantity, stop reason, fault photo	No plan override or cost access
Stores	Expected deliveries, PO balance, bin/batch state	Scan PO; enter lot/serial; damage photo; GRN draft	Final GRN and stock posting
Quality	Inspection queue, limits, sampling, prior readings	Capture readings/photos; complete inspection draft	Release, scrap, return or stock movement
Maintenance	Assigned work, asset history, downtime, checklist, spares	Accept job; readings; labour; photos; hold/complete draft	Safety handback and controlled close

Safe scan sequence



Machine safety boundary
XELOR Pocket may show accepted equipment evidence and help record maintenance work. It must not control a PLC, robot, interlock or safety function from a general mobile dashboard.

Capture media safely

Use photos for damaged material, packaging, faults, repairs, inspections and certificates. Store files in controlled object storage with tenant/transaction metadata, hash, MIME and size validation, malware scan, uploader, capture time, retention and expiring authorised links. OCR and AI create reviewable drafts only.

13. Alerts, approvals and offline boundaries

Mobile trust depends on clear severity, evidence before action, consistent acknowledgement and honest offline states.

Alert vocabulary

Level	Meaning	Mobile behaviour	Example
Critical	Already failed, stopped or breached	Immediate push, persistent inbox and escalation	Production-impacting machine down
Urgent	Likely to stop or become late today	Working-hours push and role escalation	Late material threatens today's job
Attention	Review before it becomes urgent	Inbox and digest, normally no interruption	Quote or certificate nearing expiry

Notification lifecycle: **Seen** -> **Acknowledged** -> **Resolved**. Store it server-side so phone and desktop agree. Push payloads contain minimal lock-screen information and an authenticated deep link; prices, PII and security details appear only after login.

Offline capability classes

Class	Allowed offline	Never claim
Read snapshot	Last authorised dashboard, instructions and recent history with visible as-of time	That cached data is live
Capture draft	Fault note, photos, readings, production counts and GRN draft	That the business transaction posted
Authoritative commit	Not offline; wait for server validation	Approval, stock movement, payment, quality release or award

- Every queued operation gets a stable operation ID and idempotency key.
- The device records capture time; the server records authoritative commit time.
- On sync the server rechecks permission, tenant, plant, document version, quantities and workflow state.
- Conflicts return to a person; never use silent last-write-wins.
- Cache fields are allow-listed and keyed by tenant plus user; responses are no-store by default. Never cache bank data, drawings, margins, full BOMs or personal data.
- Snapshots and drafts have a short policy TTL, no longer than the relevant shift by default. Shared-device handoff clears that user's caches and drafts.
- Use a device-bound WebCrypto key where available, attachment size limits and resumable upload. Encryption reduces casual exposure but cannot defeat active XSS on an unlocked session.
- Revocation blocks future server access and triggers best-effort purge when the app reconnects; it cannot instantly erase a disconnected lost device.
- A successful offline save says **Saved as draft**, never **Posted**.

14. Mobile security and experience standards

Factory phones may be personal, shared, low-end, intermittently connected and used near machinery. The design must account for that reality.

Security before production

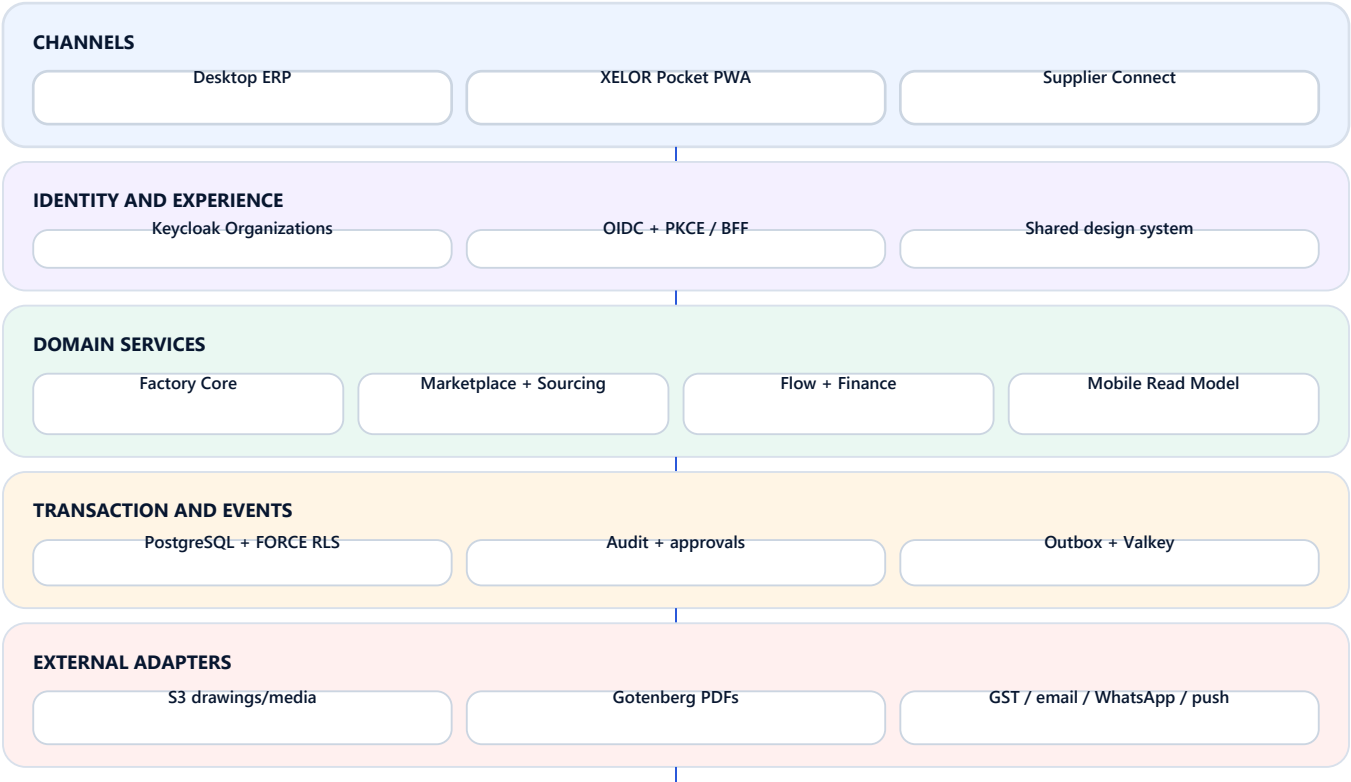
- Move OIDC access and refresh tokens out of browser-readable session storage into a same-origin BFF session with Secure, HttpOnly and SameSite cookies.
- Protect cookie-authenticated mutations with strict same-origin rules, Origin and Sec-Fetch-Site validation, and a per-session CSRF token; rotate sessions and invalidate them on logout.
- Use authorization code with PKCE; require MFA/passkeys and recent step-up for privileged awards, payments and administration.
- Use separate employee and supplier audiences/scopes. Maintain a device/session registry, remote revoke and best-effort cache purge behavior.
- Support personal-device and shared-shift-device modes with different idle locks and retained state.
- Keep lock-screen notifications generic and exclude employee PII, customer detail, prices and security evidence.
- Record scan, capture, upload, sync, decision and correction events in the audit trail.
- Use narrowly scoped, user-initiated camera permission; do not turn geolocation into employee surveillance.

Mobile UX rules

Rule	Design implication
Task-first	Say Receive material, not open the full Purchase module
One-thumb	Primary controls sit within easy reach and remain stable
Large targets	At least 44-48 px, larger for gloved shop-floor work
No wide tables	Use ranked lists, summaries and drill-down detail
Explicit state	Queued, Draft, Posted, Rejected and Offline use text plus icon
Low-end performance	Restrained motion, small payloads and progressive media upload
Local language readiness	English first, then pilot language; keep item codes and units exact
Evidence before decision	Every approval opens facts and source records first
Freshness	Every cached or delayed number shows last update

15. Target technical architecture

Extend the current Next.js, NestJS and PostgreSQL modular monolith. Do not split into microservices or external search infrastructure before measured scale requires it.



Key implementation decisions

- Dedicated mobile route group and aggregated `/mobile/*` APIs so Home does not require 15-20 calls.
- Installable PWA manifest, service worker, web push and encrypted short-lived draft storage.
- Event-built mobile read model with versioned events, idempotent consumers, projection watermark, lag threshold, dead-letter handling, replay/rebuild and periodic reconciliation. Approvals reread the source document.
- PostgreSQL full-text search and `pg_trgm` for the first supplier catalogue; add a search service only after latency and scale evidence.
- Transactional outbox and Valkey/BullMQ for notifications, projections and supplier performance.
- Encrypted object storage for drawings, photos and certificates; Gotenberg for controlled commercial PDFs.
- Keycloak Organizations before external supplier production use.
- Marketplace owns publication and verification; Sourcing owns RFQ, quote and award; Planning exposes requisitions; Purchase owns PO/GRN and the one-time award conversion port; Quality emits inspection facts.
- The mobile and supplier layers call existing domain services; they do not recreate stock, tax, quality or approval rules.

Evidence: [XE-3], [XE-4], [XE-5].

16. Data ownership, marketplace isolation and trust

Do not combine every factory's private database. Publish a sanitized projection and share transaction data only with explicit participants.

Zone	Examples	Visibility
Private tenant	BOMs, stock, costs, customers, MRP and internal vendor notes	One buyer company only
Supplier private	Contacts, bank details, unpublished capacity and documents	Supplier plus authorised verification staff
Published network	Allow-listed <code>supplier_public_snapshot</code> with profile, capability, listing and evidence status	Cross-tenant read; supplier-owner write
Participant transaction	Immutable <code>rfq_shared_revision</code> , attachment grant, clarification and award	Buyer plus the named invited supplier
Supplier-specific	One supplier's quote and revisions	Buyer and that supplier; never competitors
Trust evidence	GRN punctuality, accepted/rejected quantity and dispute outcome	Raw evidence restricted; aggregate only after sample threshold

Core new entities

Domain	Entities
Supplier network	<code>supplier_org</code> , <code>supplier_public_snapshot</code> , <code>location</code> , <code>capability</code> , <code>listing</code> , <code>certification</code> , <code>verification</code> , <code>profile_revision</code>
Sourcing	<code>rfq_private</code> , <code>rfq_shared_revision</code> , <code>attachment_grant</code> , <code>invitation</code> , <code>quote_revision</code> , <code>quote_line</code> , <code>comparison_snapshot</code> , <code>award</code>
Tenant relationship	<code>vendor_marketplace_link</code> , <code>approved_supplier_item</code> , <code>supplier_score_period</code> , <code>supplier_risk_event</code>
Commercial	<code>sales_quote_revision</code> , <code>supplier_invoice</code> , <code>three_way_match</code> , <code>payment_proposal</code> , <code>settlement_allocation</code>
Mobile	<code>device_session</code> , <code>notification_recipient</code> , <code>acknowledgement</code> , <code>scan_alias</code> , <code>sync_operation</code> , <code>media_object</code>

Enforceable database boundary

Keep restrictive equality-based tenant RLS on private ERP and `rfq_private` tables. RLS controls rows, not columns, so supplier-visible fields live in a separate immutable `rfq_shared_revision`. Each invitation/package is keyed by `buyer_org_id`, `participant_org_id`, `rfq_id` and revision; attachment ACLs are separate. A supplier session sets `actor_org_id`, reads only its matching package and writes quote rows constrained by a composite invitation key. `supplier_public_snapshot` is the only cross-tenant discovery surface. Test buyer, invited supplier, uninvited supplier, competitor and support roles. Never use `BYPASSRLS` or database-owner authority.

Lifecycle and privacy

Store purpose, source, consent, `published_by`, `verified_at`, `expires_at` and withdrawal state. Remove a withdrawn listing from discovery immediately while retaining governed transaction/audit records under the retention and legal-hold policy. Encrypt and mask bank/contact fields, log reasoned access, support correction and takedown, and define supplier suspension and appeal.

17. Repository-grounded module changes

The implementation should extend current ownership boundaries and preserve the working transaction invariants.

Module	Existing foundation	Required extension
Engineering	Item and BOM master	Category attributes, approved equivalents, listing map and drawing confidentiality
Planning	MRP, pegging and purchase requisitions	Expose requisition port/event; consume sourcing and arrival status
Marketplace (new)	Data import and vendor evidence patterns	Published supplier snapshot, capability, catalogue, consent and verification
Sourcing (new)	Purchase requisitions and workflow platform	RFQ, quote revisions, comparison, clarification, award and acknowledgement
Purchase	Vendor, PO, approval and GRN	Vendor-marketplace link and unique award-to-PO conversion port
Quality	Inspections, findings and CAPA	Emit immutable incoming-quality facts; certificate validity and dispute evidence
Inventory	Warehouses, balances and append-only ledger	Barcode receive/count, package identity and offline draft capture
Sales + Accounts	Order, dispatch, GL, AR invoice and receipt	Quotation plus AP invoice, match, payment, settlement and reconciliation
Production + Maintenance	Orders, operations, assets, downtime and work orders	Pocket queues, QR flows, capture, evidence and escalation
Integration + Import	Statutory flows, webhooks, outbox and controlled files	GST production adapters, messaging/push, supplier callbacks and catalogue import
Administration	Roles, permissions, audit and privacy	Supplier consent, publication policy, device registry and participant access

Important current gaps to close honestly

- Sales quotation and revision lifecycle is absent and documented as a major commercial gap.
- Supplier invoice, three-way match and direct-material AP are not complete.
- AP delivery also needs duplicate/tax controls, protected bank masters, maker-checker payment, settlement and reconciliation.
- Branded PDFs and sharing need production Gutenberg/object-storage integrations.
- Statutory e-invoice/e-way results are currently simulated and must not be presented as production filing.
- The responsive shell has no installable PWA, service worker or durable offline queue.
- Factory Connect is simulator evidence today; phone screens must not claim live telemetry without a real connector.

Evidence: Repository anchors: [XE-1] through [XE-8].

18. Phase 0: factory groundwork and production foundations

Duration: 4-6 weeks. Field operations are part of product development, not a sales activity postponed until after software is built.

Field programme

Workstream	Target	Output
Buyer discovery	10-15 factories in one cluster	Recurring shortages, current sourcing flow, approval rules and mobile questions
Supplier onboarding	30-50 suppliers in 2-3 categories	Capability, machinery, capacity, certificates, service radius and consent
Real RFQs	At least 5 buyer requirements	Three comparable responses for each qualified requirement
Mobile observation	Owner, buyer, stores, supervisor, operator, quality and maintenance	Five highest-frequency phone jobs, device and network matrix
Data policy	Buyer, supplier and legal review	Publication consent, confidentiality, retention and verification policy

Platform foundations

- Define the Keycloak Organizations mapping, employee/supplier audiences and supplier principal types; prove them in an isolated pilot realm.
- Spike encrypted object storage, Gotenberg, observability and malware scanning; production hardening remains a Phase 1 external-pilot gate.
- Complete the participant threat model, permission registry and negative cross-participant test design.
- Prototype mobile role dashboards at 360-430 px and validate them in real factory lighting and connectivity.
- Define supplier category taxonomy, required attributes and verification evidence.
- Capture baseline cycle times, quote liquidity, paper delay, approval delay and data quality.

Exit gate

Five real buyer needs in the discovery sample each receive comparable responses from at least three qualified suppliers; the mobile role map, security model, pilot buyers and supplier cohort are committed. This five-of-five learning gate is deliberately stricter than the later operating KPI.

19. Delivery phases

The sequence productionises one combined proof slice first, then deepens sourcing evidence, finance and safe mobile work before network scale.

Phase	Indicative duration	Primary scope	Exit gate
1. Combined pilot slice	12-14 weeks	Sales quotation/PDF; Pocket pulse; supplier basics; requisition-to-RFQ; quote revisions; comparison; award-to-PO; production security foundations	Enquiry reaches order; real shortage reaches approved draft PO; owner sees it on phone
2. Source operational depth	10-12 weeks	Catalogue search; secure files; clarification; acknowledgement; GRN/quality events; performance facts and insufficient-evidence scorecard	Real shortage reaches PO, GRN and quality evidence with no retyping
3. Finance completion + Mobile Work	10-14 weeks	Supplier invoice; duplicate/tax/match controls; AP; maker-checker payment/settlement; scans; inspections; production/maintenance queues; offline drafts	PO-to-settled-payment record and selected floor tasks complete in pilot
4. Trust and network scale	12-16 weeks after evidence	Richer search; certificate expiry; capacity; score publication after sample threshold; disputes; regional language; APIs; multi-site sourcing	First cluster meets liquidity, quality and repeat-use thresholds
5. Ecosystem	Evidence-led	Image/spec extraction; explainable matching; partner logistics/payment protection; native app decision	Only after data, economics and user evidence justify it

Priority matrix

P0 means mandatory before general availability of XELOR Unified, not all delivered in the first phase. Phase exit gates still govern sequence.

P0 - must build	P1 - should build	P2 - later
Sales quotation and quote-to-order	Mobile receiving and inspection	Open public bidding
Supplier verification and structured RFQ	Barcode/QR and offline drafts	Image-assisted part search
Revised quotes and landed comparison	Catalogue bulk import and repeat RFQ	Managed logistics or escrow
Governed award-to-PO	WhatsApp/email document delivery	Lending or invoice finance
Pocket pulse, alerts and approvals	Capacity calendar and regional language	Native apps and national rollout
Supplier invoice and three-way match	Buyer-configured score weights	Public anonymous reviews

20. Mobile solution phase in detail

Mobile delivery should grow from visibility to controlled action, then to supplier participation and hardened intermittent-connectivity support.

Mobile phase	Scope	Who benefits	Acceptance
M0. Groundwork	Observe shifts, devices, network, shared use, labels and five key jobs	All pilot roles	Journey, threat and device matrix approved
M1. Visibility	Installable PWA, Home, My Work, alert inbox, evidence deep links and freshness	Owner, plant head, purchase	80% of five core scenarios answered without desktop/calls; freshness understood
M2. Approvals + capture	Step-up, scan, production counts, maintenance, inspection, GRN draft, media	Supervisor, operator, stores, quality, maintenance	At least 90% success on selected tasks; zero duplicate commits; audit complete
M3. Marketplace mobile	RFQ review, supplier portal, quote revisions, clarifications, shortlist, PO acknowledgement	Buyer and supplier	Supported RFQ response and award-review tasks complete without desktop; permission tests pass
M4. Hardened offline	Encrypted queue, resumable media, conflicts, shared device, revoke and multilingual pilot	Low-connectivity factories	99% eligible queued sync; injected conflicts resolved; revoke/logout tests pass

Actions that remain desktop-led

- BOM engineering, drawing release and complex product configuration.
- Finite planning, large quote comparison and multi-line exception analysis.
- Role design, segregation-of-duties configuration and bulk administration.
- Accounting period close, large reconciliations and statutory configuration.
- AI provider configuration, security investigation and complex data imports.

Phone promise

Not everything is editable on a phone. Everything relevant to a user's responsibility should be visible, fresh, understandable and linked to a safe next action.

21. Delivery team, sequencing and operating model

A stable 10-12 person product/engineering team plus field onboarding can deliver the first combined pilot if ownership remains clear and phase gates prevent scope expansion.

Indicative MVP team

Role	Count	Primary accountability
Product lead / manufacturing procurement owner	1	Outcome, scope, pilot and workflow truth
Technical lead / architect	1	Domain boundaries, invariants and integration
Backend engineers	3	Sourcing, commercial, mobile APIs and participant security
Frontend/PWA engineers	2	Desktop extensions, Pocket and Supplier Connect
Product designer / field researcher	1	Role journeys, factory observation and usability
QA automation	1	E2E, permission, sync and regression coverage
Platform/SRE/security	1	Identity, storage, observability, delivery and threat controls
Data/search engineer	0.5-1	Taxonomy, search, normalisation and score processing
GST/compliance adviser	Fractional	Document and statutory review
Field onboarding	2-4	Buyer/supplier data, verification and pilot success

Critical sequencing

- 1 Identity, participant authorization and data classification before supplier portal access.
- 2 Object storage security before drawings, certificates or phone photos.
- 3 Published supplier projection before cross-tenant search.
- 4 RFQ/quote state machine and immutable revisions before comparison UI.
- 5 Approval and idempotent award conversion before live procurement use.
- 6 Transaction history before supplier scoring or AI recommendations.
- 7 Mobile read APIs before dashboards; server validation before offline writes.
- 8 Observability and leak probes before external pilot.
- 9 PWA telemetry before funding native applications.

22. Pilot scorecard

Targets are hypotheses for the first cluster. Phase 0 establishes baselines and adjusts targets before commitments.

Area	Measures	Initial target hypothesis
Marketplace liquidity	Qualified RFQs with 3 quotes; first response; award rate; repeat usage	70% with 3 compliant quotes in 48h; first compliant quote under 8 business hours
Procurement	Shortage-to-RFQ; RFQ-to-PO; approval; OTIF; incoming acceptance	30% cycle reduction; every marketplace PO captures OTIF and inspection
Commercial	Quote preparation; quote-to-order; dispatch-to-invoice; ageing	Configured quote under 5 minutes; 100% quote-to-order trace
Mobile adoption	Weekly active users; task completion; approval; alert response	70% weekly active targeted managers; 60% mobile approval completion
Mobile performance	Interactive time, crash-free sessions, sync and overflow	p95 under 2.5s on representative 4G; 99% sync; zero horizontal overflow
Mobile task quality	Success, error and time by operator, stores, quality, maintenance, buyer and supplier; conflict recovery; shared-device logout	At least 90% success in selected pilot tasks; zero duplicate commits; all revoke/logout tests pass
Factory capture	Point-of-work digital events and paper delay	80% of selected events digital; paper delay reduced 70%
Data trust	Catalogue freshness, stale tiles, alert false positives, first-pass match and disputes	Baselines in Phase 0; every stale tile visible; every dispute attributable
Governance	Award trace, file audit, idempotency, leakage, negative access tests and verification expiry	100% trace/audit/idempotency/access-test coverage; zero confirmed leakage
Reliability	API, outbox, read-model lag, restore, RPO/RTO and notification delivery	99.9% core API in agreed window; outbox p95 under 60s; restore drill meets agreed RPO/RTO

Measurement contract

Before pilot commitment, define qualified RFQ, compliant quote, business-hour calendar, OTIF, sync-success denominator, availability window, representative device/network, alert false positive and first-pass match. Record event start/end, cohort, exclusions, measurement window and metric owner. The Phase 0 five-of-five response gate is a small discovery test; 70% is the later rolling operating hypothesis.

North-star metric

Percentage of purchase shortages that become an approved, on-time and quality-accepted supply outcome with no manual re-entry between requirement, quote, PO, receipt and inspection.

23. Principal risks and controls

The combined solution adds cross-company data, public onboarding, phone access and intermittent connectivity. Those are product risks, not later infrastructure details.

Risk	Failure mode	Control
Cold start	No buyer/supplier liquidity	Anchor buyers, one cluster, repeat-demand categories and supplier invitations
Catalogue quality	Wrong specification, stale price or duplicate item	Category schema, UOM normalisation, moderation, deduplication and freshness expiry
Fake verification	Badge implies evidence that was never checked	Evidence-specific statuses, expiry and factory-visit record
Drawing leakage	Supplier or outsider gets confidential IP	Separate shared revision, attachment ACL, short link, watermark, audit, expiry and NDA state
Quote leakage	Supplier sees competitor pricing	Supplier-specific RLS and negative contract tests
Bad award	Cheapest quote wins or conflict is hidden	Technical pass gate, raw facts, separate approver, conflict declaration and one-time PO constraint
Mobile loss	Exposed data on a lost/shared device	Short TTL, allow-listed cache, idle lock, best-effort revoke/purge and generic notifications
Offline duplication	Retry posts stock, award or payment twice	Draft-only offline, operation ID, idempotency and unique constraints
Stale status	Old telemetry appears live	Mandatory timestamp and fail-to-Unknown
Publication/privacy	Profile or contact used without valid purpose/consent	Purpose/consent register, masked contacts, withdrawal, rights handling, retention and legal hold
Network abuse	Bulk harvesting, fake RFQs or contact spam	Rate limits, bot checks, enquiry quotas, enumeration controls, consent and abuse reporting
Verification liability	Incorrect badge harms buyer or supplier	Evidence wording, expiry, correction/takedown SLA, reviewer audit and limitation policy
Dispute/suspension	Unfair score, takedown or account suspension	Raw evidence, insufficient-evidence state, correction, appeal and accountable decision
Payment fraud	Duplicate invoice or malicious bank change	Duplicate/tax/match controls, bank-change cooling, maker-checker, signed handoff and reconciliation
Messaging consent	WhatsApp delivery becomes unsolicited	Approved templates, recorded opt-in, purpose limit, opt-out and channel preference
Scope explosion	Marketplace, ERP, mobile and finance stall together	Phase gates, explicit deferred list and one transactional owner
Supplier adoption	Supplier refuses full ERP setup	Lightweight portal, mobile deep links, Excel import and no ERP requirement
AI overreach	Model publishes, awards or posts	Draft/explain only; deterministic rules and human gates

24. Non-goals and decision guardrails

Explicit boundaries protect the pilot from becoming an unfinishable national marketplace, a retail product and a factory-control system at the same time.

No data scraping Do not scrape, copy or republish IndiaMART's supplier database. Onboard with consent, imports and permitted integrations.	No private data publication Existing vendors, BOMs, costs, stock, customers and production remain private unless explicitly published through a projection.	No nationwide launch Prove liquidity and transaction outcomes in one cluster and narrow categories first.
No autonomous commerce AI cannot publish RFQs, award suppliers, create commitments, release quality or move money without rules and approval.	No complete offline ERP Offline stores snapshots and drafts. Stock, financial and controlled approvals require the server.	No mobile machine control The phone is not a PLC, interlock, robot controller or safety device.
No escrow/lending in MVP Use partners later only after legal, operational and economic evidence.	No native-app assumption Ship PWA, measure limitations, then decide whether native investment is justified.	No generic retail pivot POS and thermal breadth belong in a separate SKU only if XELOR intentionally targets traders.

Decision tests for every proposed feature

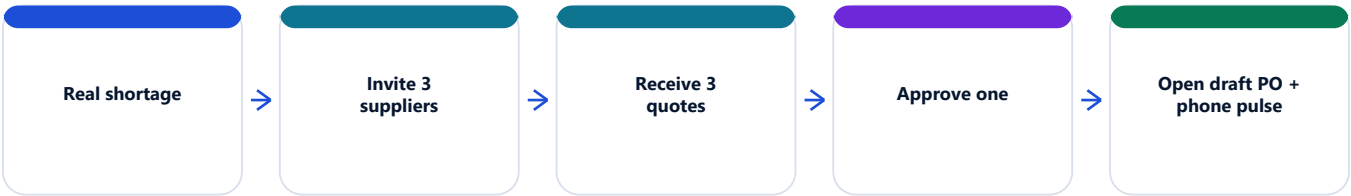
- 1 Does it reduce time or risk in a real factory or sourcing workflow?
- 2 Can it reuse the authoritative XELOR domain service rather than duplicate rules?
- 3 Is the owner, participant and visibility of its data explicit?
- 4 Can the action be audited, reversed or corrected through an owned process?
- 5 Does the phone need to act, or only see and route the task?
- 6 Is it required for the next phase exit gate, or should it be deferred?

25. First 90 days

The first quarter spans Phase 0 discovery plus a non-production engineering proof slice. It creates field evidence and closes critical design decisions; it is not a production release.

Window	Business and field	Product and design	Engineering and controls
Days 1-30	Select cluster/categories; recruit 5 buyers and initial suppliers; observe shifts; collect baseline	Journey maps; category templates; owner/plant/buyer mobile prototype	Threat model; identity plan; object storage and permission spike
Days 31-60	Run manual RFQs; verify supplier evidence; test consent and confidentiality	Clickable Source, Supplier Connect and Pocket prototypes; quotation document design	Sales quote schema/API; supplier projection; participant RLS tests; mobile read endpoint
Days 61-90	Complete first pilot RFQ and revise policies from failures	Usability test at factory and supplier sites; approve Phase 1 backlog	Non-production slice: requisition -> RFQ -> quote -> approved draft PO; installable read-only Pocket shell

Day-90 demonstration



Do not fake the demonstration
The requirement, supplier responses, permission boundaries, approval and PO link must be real pilot records. Clearly label the slice non-production and identify every identity, storage, payment, logistics, telemetry or statutory control that still requires Phase 1 hardening.

26. Final recommendation

Build the shortest credible proof slice that connects supply, factory evidence and mobile visibility, then extend it through receipt, quality, production, billing and cash under explicit phase gates.

Proceed

Fund Phase 0 and Phase 1 around one buyer cluster, one controlled supplier cohort and the combined shortage-to-approved-PO plus mobile factory-pulse proof slice. Do not call it a complete loop until receipt, quality, production, billing and cash are demonstrated.

The product in one sentence

XELOR is the industrial operating system that helps a manufacturer find qualified supply, run the factory, bill and control cash, and understand what needs attention from a phone - while keeping private data isolated and every material decision governed.

The operating loop



Why this wins

- Supplier breadth becomes useful because it begins with a real factory requirement.
- Simple mobile access becomes trustworthy because it reads the same governed records as desktop.
- Commercial convenience becomes defensible because documents connect to planning, stock, quality and ledger evidence.
- Supplier trust improves over time because ratings come from actual delivery and inspection outcomes.
- XELOR remains differentiated from marketplaces, billing tools and generic ERPs by closing the industrial loop.

Appendix A. Proposed screen inventory

This is a product inventory for estimation, not a commitment to ship every screen in the first release.

Surface	Proposed screens
Source - buyer	Supplier search; supplier profile; requisition-to-RFQ; RFQ detail; invitations; clarification; comparison; award review; sourcing history
Supplier Connect	Onboarding; verification; company profile; capability; catalogue; RFQ inbox; quote editor; quote history; PO acknowledgement; shipment update; performance
Flow - sales	Enquiry; quotation list; quote editor; revision compare; acceptance; PDF/share; order conversion; invoice register; reminder worklist
Flow - purchase/AP	Supplier invoice inbox; duplicate/tax checks; match workbench; AP ageing; bank-change review; maker-checker payment; settlement allocation; reconciliation; debit note; vendor statement
Pocket - common	Home; My Work; Scan; Alerts; More; evidence detail; approval detail; sync queue; device/security; notification settings
Pocket - production	Today queue; operation detail; start/complete; counts; blocker; handover
Pocket - stores	Expected deliveries; PO scan; receipt draft; batch/serial capture; issue material; stock lookup/count
Pocket - quality	Inspection queue; characteristic capture; evidence upload; completion; hold/disposition status
Pocket - maintenance	Assigned jobs; asset scan; fault report; downtime; task checklist; spares; labour; evidence; handback
Pocket - owner/manager	Factory pulse; order risk; production; downtime; quality; supply; cash; approvals; daily digest

Shared design components

Evidence panel; freshness badge; participant badge; approval footer; secure attachment; status history; exception explanation; empty/error/offline states; scan result; draft/sync banner; role-aware action tray; searchable item/supplier picker; audit disclosure.

Appendix B. Evidence and source register

External product facts were checked against current official pages and filings on 30 August 2026. Repository statements refer to local XELOR-MVP commit d329f09be626 on the same date.

External sources

Code	Source	Type
IM-1	IndiaMART Q1 FY27 Investor Presentation	Official source
IM-2	IndiaMART FY2025-26 Integrated Annual Report	Official source
IM-3	IndiaMART Payment Protection	Official source
IM-4	IndiaMART marketplace homepage	Official source
VY-1	Vyapar India product homepage	Official source
VY-2	Vyapar plans and pricing	Official source
VY-3	Vyapar manufacturing product page	Official source
VY-4	Vyapar Android listing	Official source
VY-5	Vyapar offline billing guide	Official source
VY-6	Vyapar quotation guide	Official source

XELOR repository evidence

Code	Repository reference	Purpose
XE-1	<code>docs/00-project/02-complete-project-context-and-business-model-input.md</code>	Product context and target customer
XE-2	<code>docs/02-investor-demo/02-capability-gaps.md</code>	Quotation, AP and production-readiness gaps
XE-3	<code>docs/00-project/01-technology-stack.md</code>	Architecture, RLS, outbox and planned production services
XE-4	<code>README.md:226,1306</code>	FORCE RLS posture, permission truth and documented boundaries
XE-5	<code>packages/db/src/client.ts:25-38; schema/engineering.ts:14; planning.ts:229,452; purchase.ts:22,40,87</code>	Tenant fence plus item/BOM, MRP, requisition, vendor, PO and GRN anchors
XE-6	<code>schema/sales.ts:39,116; accounts.ts:99,130; platform.ts:30,46</code>	Sales order, dispatch, AR, receipts, outbox and audit anchors
XE-7	<code>apps/web/src/spine/auth/session.tsx:20-25; shell/app-shell.tsx:63; app/globals.css:500; modules/registry.ts</code>	Current token storage, responsive shell and module registry
XE-8	<code>docs/01-agent-os/06-factory-connect.md:9-16</code>	Simulator boundary and telemetry truth

Interpretation note

IndiaMART and Vyapar capabilities are used as design inspiration. This blueprint does not imply permission to copy their data, branding, proprietary implementation or customer relationships.